1.1 From Bitcoin to blockchain

The initial excitement about blockchain technology was about enabling peer-to-peer transfers of digital currency to anybody in the world, crossing human-created boundaries (such as the borders of countries) without any intermediaries such as banks. This excitement was further heightened by the realization that this peer-to-peer capability could be applied to other, non-cryptocurrency types of transactions. These transactions involve assets such as titles, deeds, music and art, secret codes, contracts between businesses, autonomous driver decisions, and artifacts resulting from many everyday human endeavors. A transaction record may contain other details based on the blockchain protocol and the application.

> **DEFINITION** A *transaction* recorded on a blockchain contains a peer-to-peer message that specifies the operations executed, data parameters used for the execution of operations, the sender and receiver of the message, the transaction fee, and the timestamp of its recording.

Bitcoin has been in operation continuously since its launch. At the time of this writing, according to Blockchain Charts (https://www.blockchain.com/en/charts), it is delivering more than 200,000 transactions per day. Following its initial success, people began to ask, “If you can transact digital currency, why not any other digital assets?” This question was answered around 2013 with the addition of an environment for code execution on another popular blockchain, Ethereum (https://ethereum.org). The innovation was that the verification, validation, and recording could be extended to other digital assets and to related transactions